

Three-Points Rule

The Three-Points Rule is a specific variant of numerical integration that uses three points within each subinterval to estimate the integral. It is less commonly used than Trapezoidal and Simpson's rules but can provide a more accurate approximation under certain conditions.

Steps:

- Similar to Trapezoidal and Simpson's rules, divide the interval $[a, b]$ into n equal subintervals, each of width $h = \frac{b-a}{n}$.
- Calculate the function values at three points within each subinterval: typically at $x_i, x_{i+0.5}$, and x_{i+1} , where $x_i = a + i \cdot h$.
- Apply the Three-Point Rule formula to each subinterval:

$$f'(x_0) \approx \frac{1}{2h} \cdot [-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)]$$

where h is the step size between the points.

- Sum up the contributions from all subintervals to obtain the numerical approximation of the integral.

Problem Approximate $f'(2)$ using the three-point rule, given the table of values:

x	2.0	2.5	3.0	3.5	4.0	4.5
$f(x)$	0.5	0.4	0.3333	0.2857	0.25	0.2222

Is possible to use the forward difference formula for the derivative at $x = 2.0$, which is:

$$f'(x_0) \approx \frac{1}{2h} \cdot [-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)]$$

Here, $h = 0.5$. To approximate $f'(2.0)$, is possible to use the points $x = 2.0$, $x = 2.5$, and $x = 3.0$. Plugging these into the formula:

$$f'(2.0) \approx \frac{-3f(2.0) + 4f(2.5) - f(3.0)}{2 \cdot 0.5}$$

Substituting the given function values:

$$f'(2.0) \approx \frac{-3(0.5) + 4(0.4) - 0.3333}{1}$$

Simplifying the numerator:

$$f'(2.0) \approx \frac{-1.5 + 1.6 - 0.3333}{1} = \frac{0.1 - 0.3333}{1} = -0.2333$$

Thus, the approximate value of $f'(2.0)$ using the three-point rule is:

$$f'(2.0) \approx -0.2333$$